BTL1213 Operating Systems / BIT1213 Operating Systems / BCL1213 Operating Systems

SEGi University

Chapter 01 Introduction to Operating Systems



BACHELOR OF INFORMATION TECHNOLOGY (HONS)

BACHELOR OF COMPUTER SCIENCE (HONS)

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Learning Objectives:

By the end of this lesson, students will be able to:

- Learn the fundamentals of operating system.
- Identify the history of operating system.
- Describe the operating system architecture.
- Outline the types of operating system.

OVERVIEW

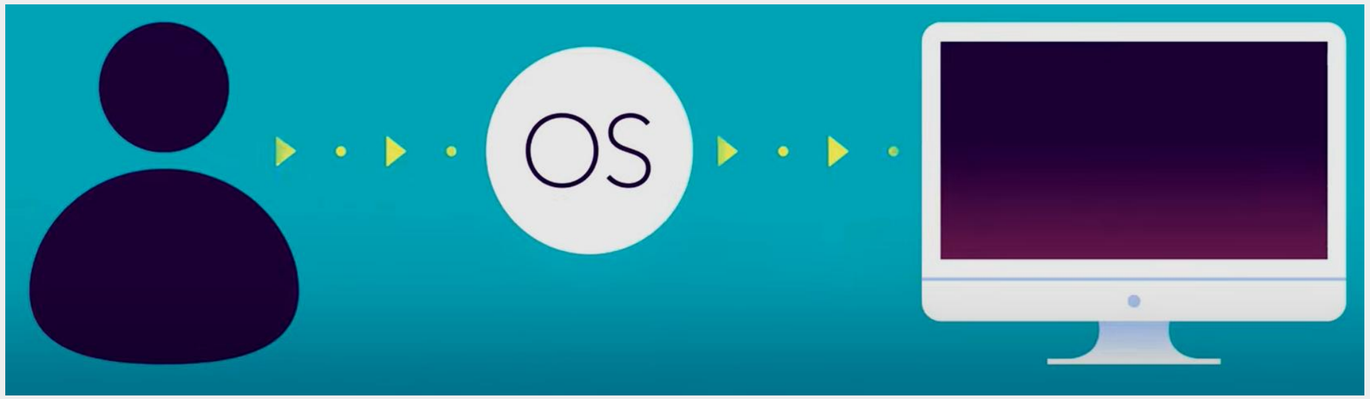
Topic for today's main discussion:

- What is Operating Systems?
- History of Operating Systems
- Operating Systems Architecture
- Types of Operating Systems






01 What is Operating Systems?



- Operating System, often called as OS, is a program (software) that let you interact with your computer (hardware).
- Also known as a **resource manager.**

- Tasks handled by OS are file management, task management,memory management, process management, disk management, I/O management, peripherals management, etc.

How OS works?

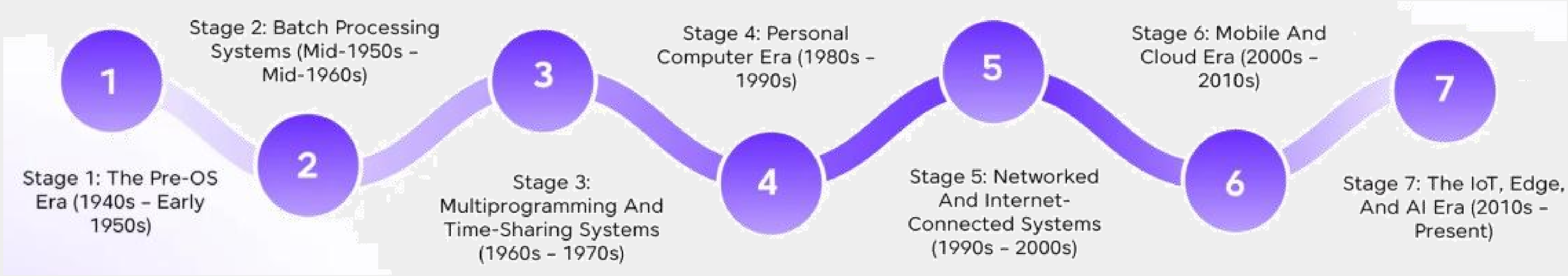
- Once installed, an operating system relies on a large library of device drivers to tailor its services to the specific hardware environment.
- For example, every application could make a common call to a storage device.
- The OS receives that call and uses the corresponding driver to translate the call into actions, or commands, needed for the underlying hardware on that specific computer.

- The operating system provides a comprehensive platform that identifies, configures and manages the following:
    - A range of hardware, including processors.
    - Memory devices and computer memory management.
    - Chipsets, storage, networking.
    - Port communication interfaces.
    - Subsystem interfaces.

Functions of OS

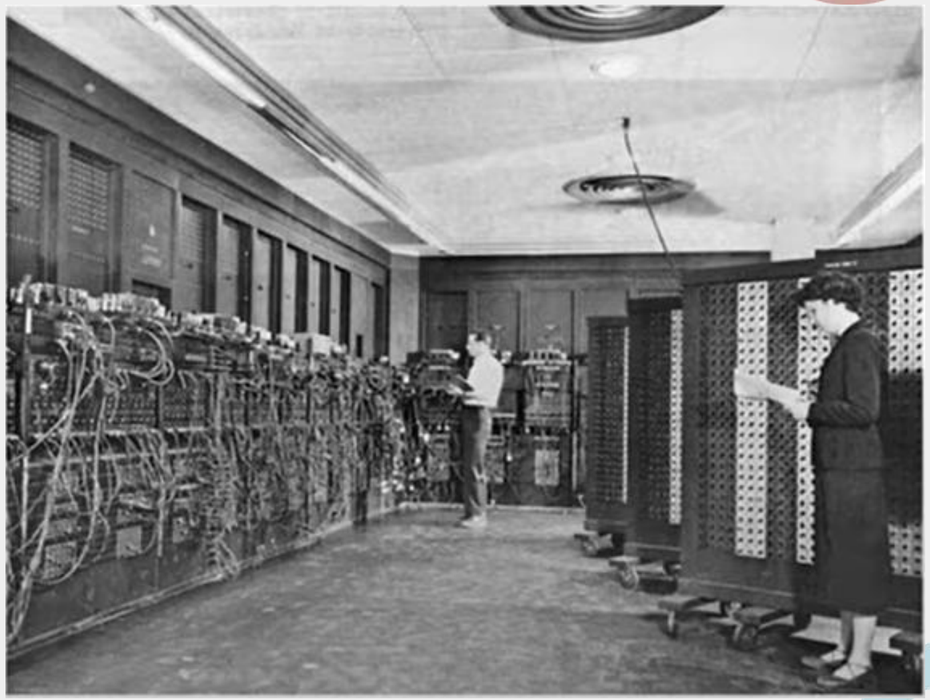
- From an end user's perspective, an operating system provides the following three essential capabilities:
  - Offers a UI through a CLI or GUI.
  - Launches applications and manages application execution.
  - Identifies and determines the allocation of system hardware resources such as printers and backup disk drives to those applications, typically through a standardized API.

02 History of Operating Systems



a. 1940s - 1950s : Early Beginnings

- Computers were programmed using thousands of different connectors; memory was a bunch of tubes with mercury inside.
- Most famous computer without OS was the U.S government’s ENIAC (Electronic Numerical Integrator and Computer).



b. 1960s : Multiprogramming and Timesharing

- Introduction of multiprogramming to utilize CPU efficiently.
- CTTS (1961) and Multics (1969), allow users to interact with a single system.

c. 1970s : UNIX (1973)

- Company Bell Labs released UNIX, the first OS that could work on different systems.
- UNIX offered simplicity, portability and multitasking.
- More impler OSs introduced, CP/M (1974) and PC-DOS (1981).



d. 1980s : GUI and Networking

- Graphical User Interfaces (GUI) gained popularity among Apple Macintosh (1984) and Microsoft Windows 1.0 (1985).
- Networking features, TCP/IP in UNIX, became essential.

e. 1990s : Linux and Advanced GUIs

- Linux (1991) introduced a free and an open-source development.



f. 2000s-Present : Mobility and Cloud Computing

Mobile OSs like iOS (2007) and Android (2008) dominate.

Cloud-based and virtualization technologies emerged, like Windows Server and Linux.

g. 2020s: AI Integration - (Ongoing)

03 Operating System Architecture

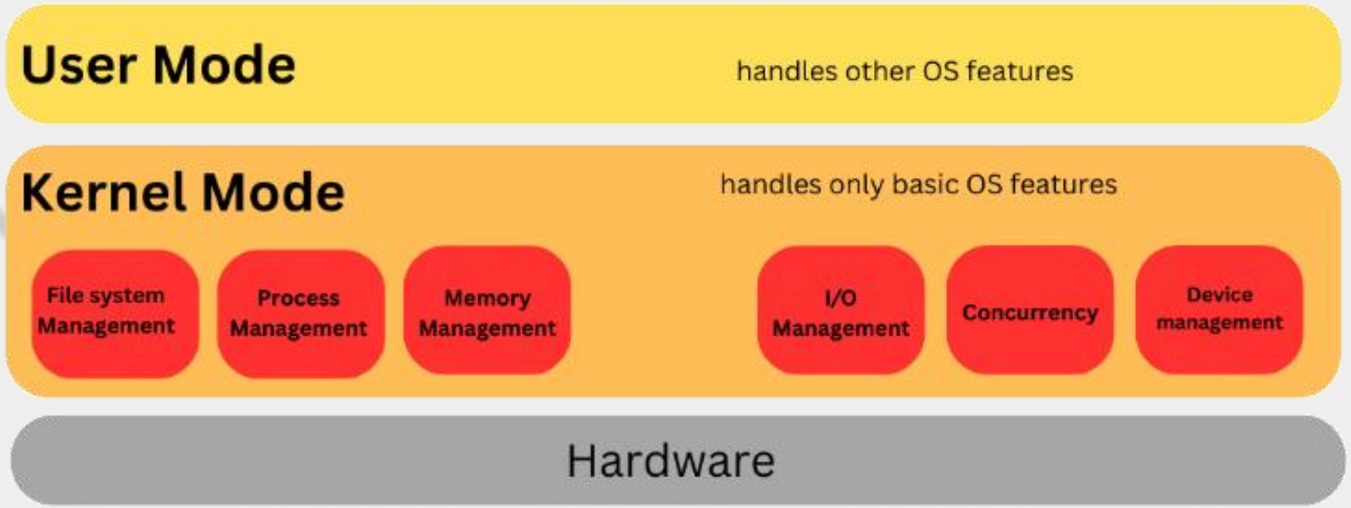
- OS Architecture deals with what the OS aims to achieve and how it interacts with external systems.
- Different architectures can impact system performance, flexibility, and security.
- Take a look at all 5 architectures.

a. Monolithic Architecture :

- Includes all system functionalities in a single module.
- The OS as a whole is working in the kernel space.
- The kernel directly controls all.
- Example : LINUX, UNIX

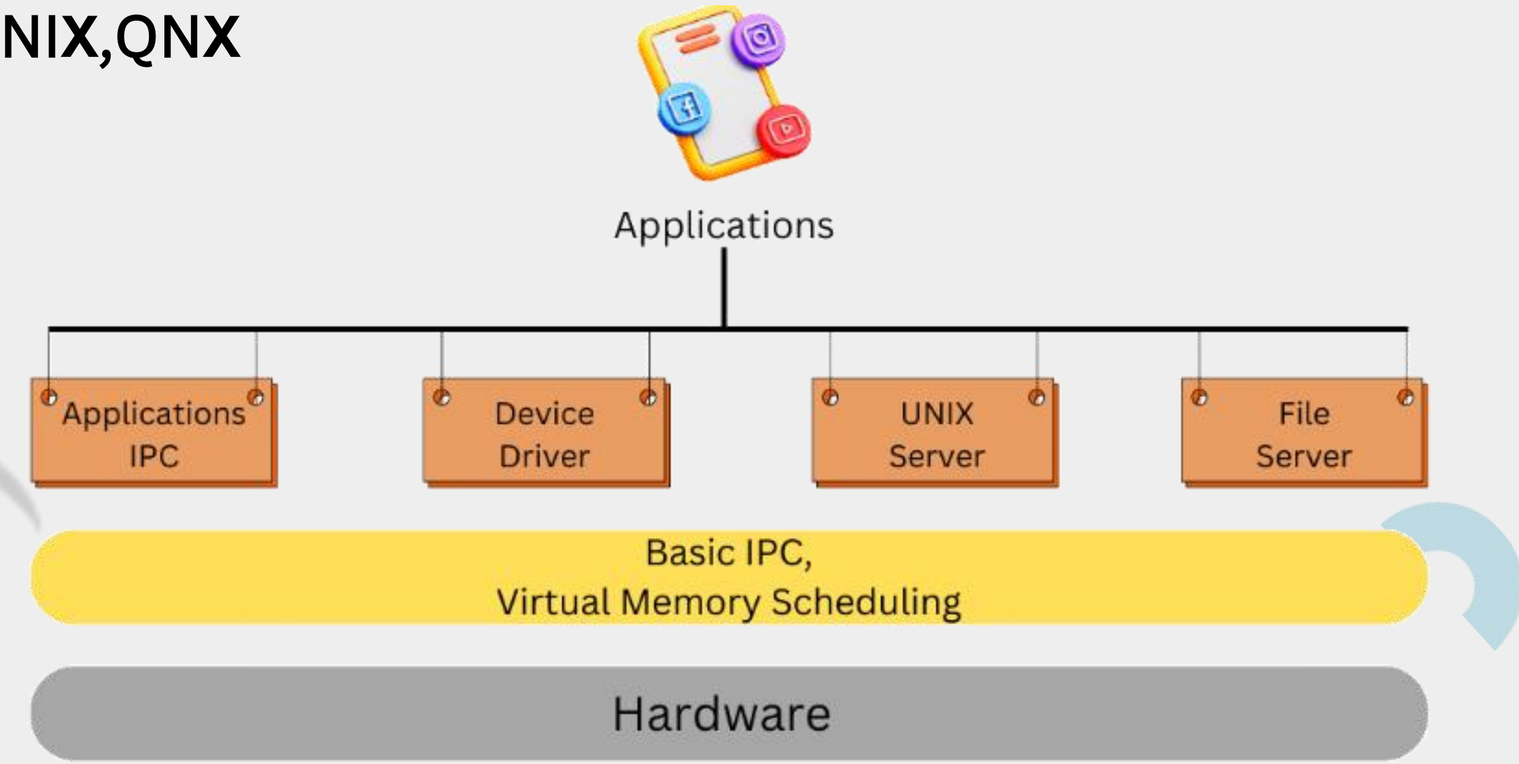


Applications



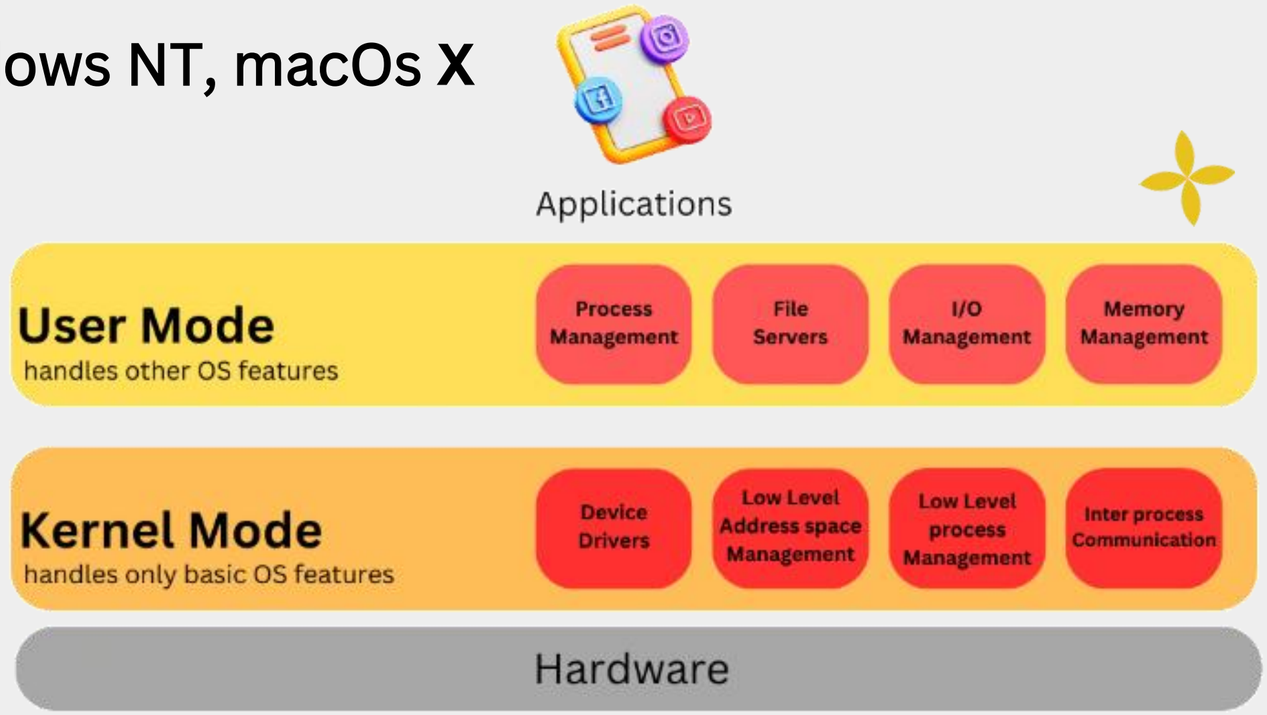
b. Microkernel Architecture :

- Includes only major functions : memory and process management in the kernel.
- File management and input-output management are included in the user space.
- Example : MINIX,QNX



c. Hybrid Architecture :

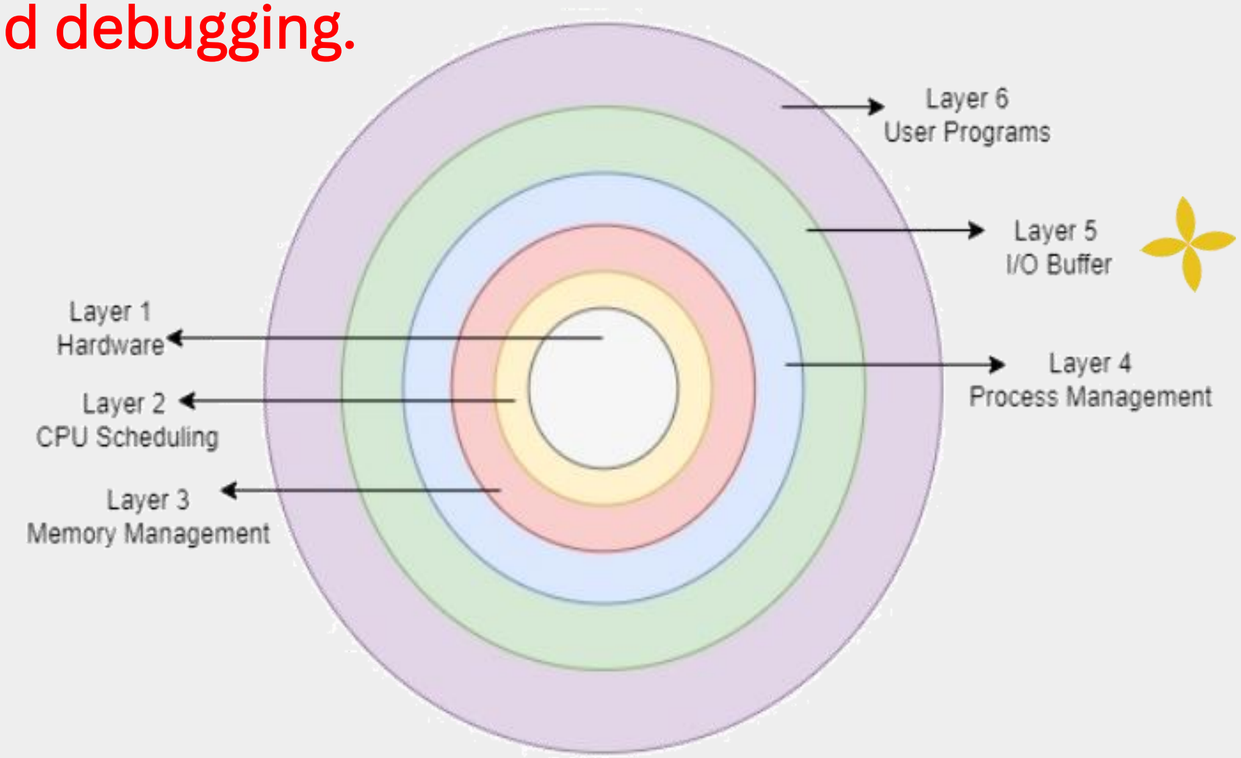
- A combined of both microkernel and monolithic kernel architectures.
- A design like the monolithic kernel. It is fast, like the monolithic kernel. Inherits the modularity and stability of microkernel architecture.
- Examples : Windows NT, macOs **X**



d. Layered Architecture

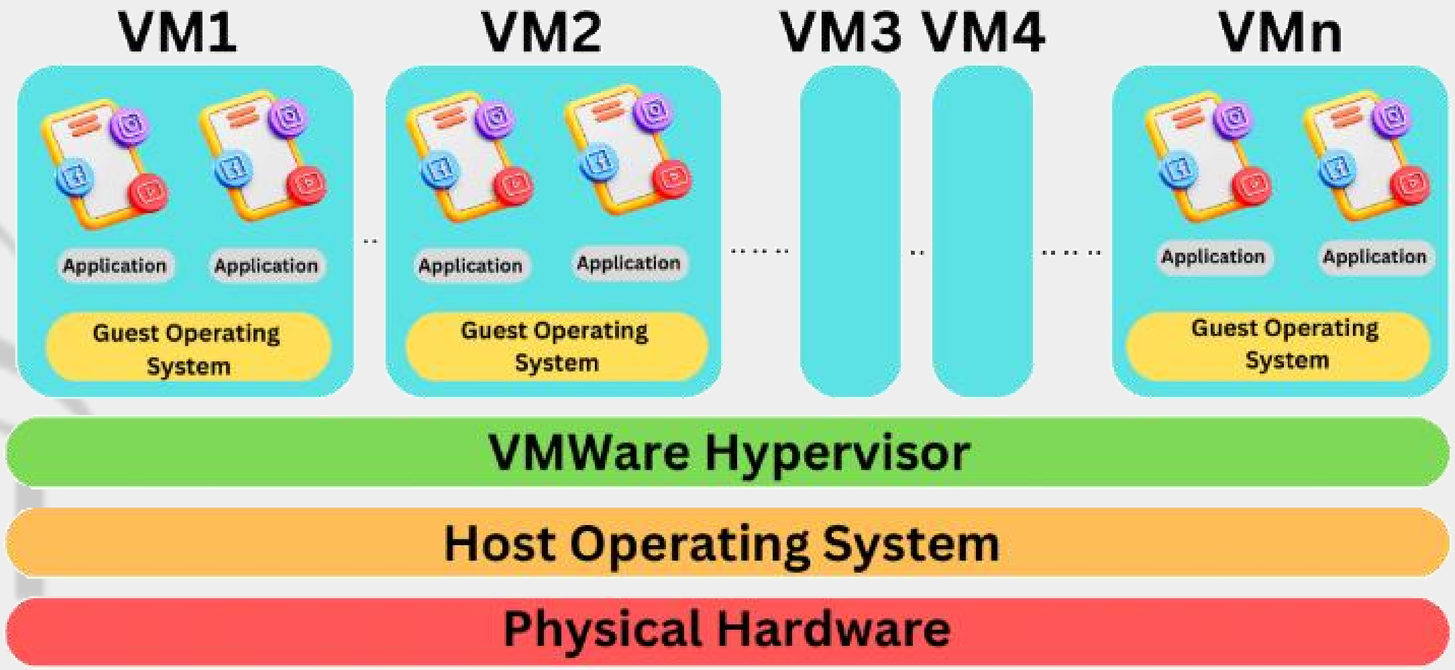
- The Layered is a way of organizing the different components of an OS into separate layers.
- Each layer is responsible for specific functions and interacts with the layer above and below it.
- Example : Modern Linux, Windows

- The bottom layer is the hardware layer, followed by the kernel layer, system call interface, and shell layer.
- This structure helps in modularizing the OS and allows for easier maintenance, development, and debugging.



e. Virtual Machine Architecture

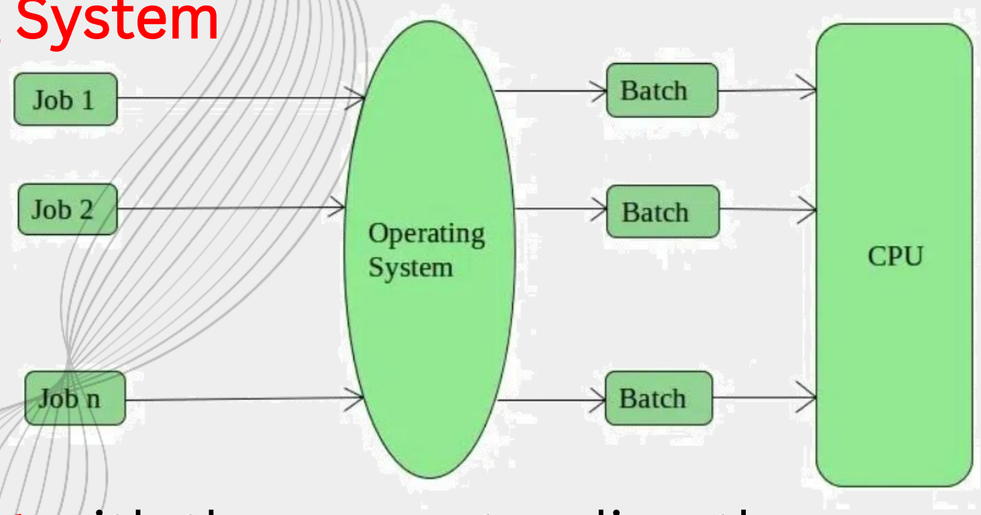
- The virtual machine architecture runs multiple operating systems on a single physical machine.
- This approach provides flexibility and efficiency.



04 Types of Operating Systems

- OS can be categorized according to different criteria whether for mobile devices (examples Android and iOS) or desktop (examples Windows, macOS and Linux).
- Here we see different types of OS.

a. Batch Operating System

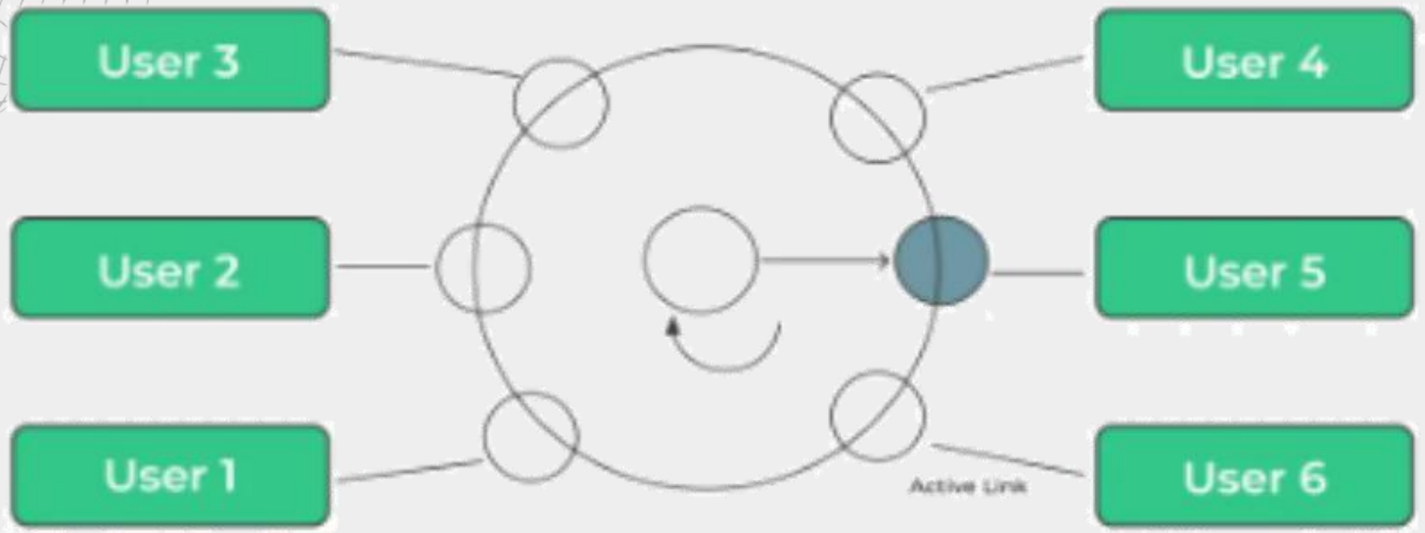


- Does **not interact** with the computer directly.
- There is an operator which takes similar jobs having the same requirements and groups them into batches.
- **To manage and execute a large number of jobs efficiently by processing them in groups.**

Advantages	Disadvantages
Many users can share batch systems. There is little idle time for batch operating systems.	Batch operating systems are challenging to debug.
It becomes possible to manage large workloads.	Any failure of the system creates a backlog.
Estimating how long a task will take to be completed is easy.	It may be costly to install and maintain sound batch operating systems.

b. Time-sharing or multitasking OS

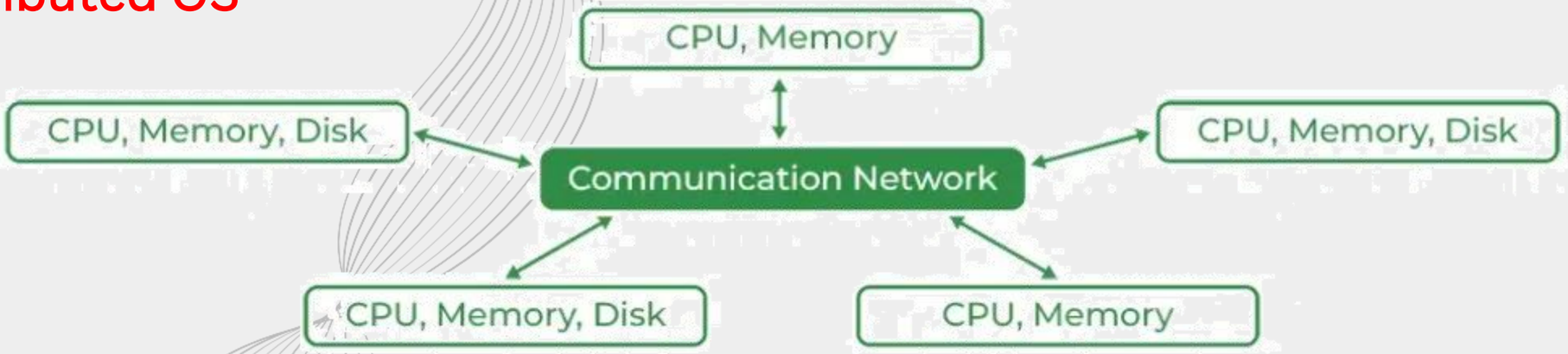
- Every process working in round robin manner and given some time to execute.
- The time that each task gets to execute is called quantum.
- After this time interval is over OS switches over to the next task.
- Examples : Multics and Unix.



Advantages	Disadvantages
There's a quick response during task performance.	The user's data security might be a problem.
It minimizes the idle time of the processor.	System failure can lead to widespread failures.
All tasks get an equal chance of being accomplished.	Problems in data communication may arise.
It reduces the chance of software duplication.	The integrity of user programs is not assured.

- Example of Time-sharing or multitasking OS : IBM VM/CMS (1972), TSO (TIME SHARING OPTION - 1960s), WINDOWS TERMINAL SERVICE

c. Distributed OS



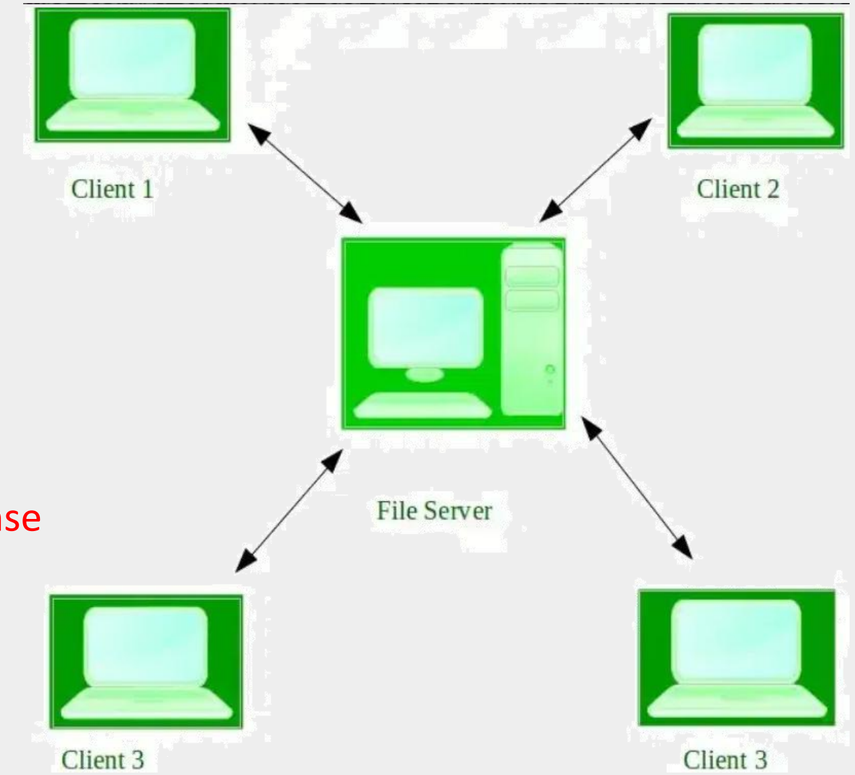
- Based on autonomous but interconnected computers communicate via lines or a shared network.
- Each autonomous has its own processor.
- Used in telecommunication networks, airline reservation control and peer-to-peer networks.

Advantages	Disadvantages
They allow remote working.	If the primary network fails, the entire system shuts down.
They allow a faster exchange of data among users.	They're expensive to install.
Failure in one site may not cause much disruption to the system.	They require a high level of expertise to maintain.

d. Network OS

- Network os are installed on a server, providing users with the capability to manage data, user groups and applications.
- To access and share files and devices, mainly in a local area network.
- Examples: Microsoft Windows, Linux and macOS X.

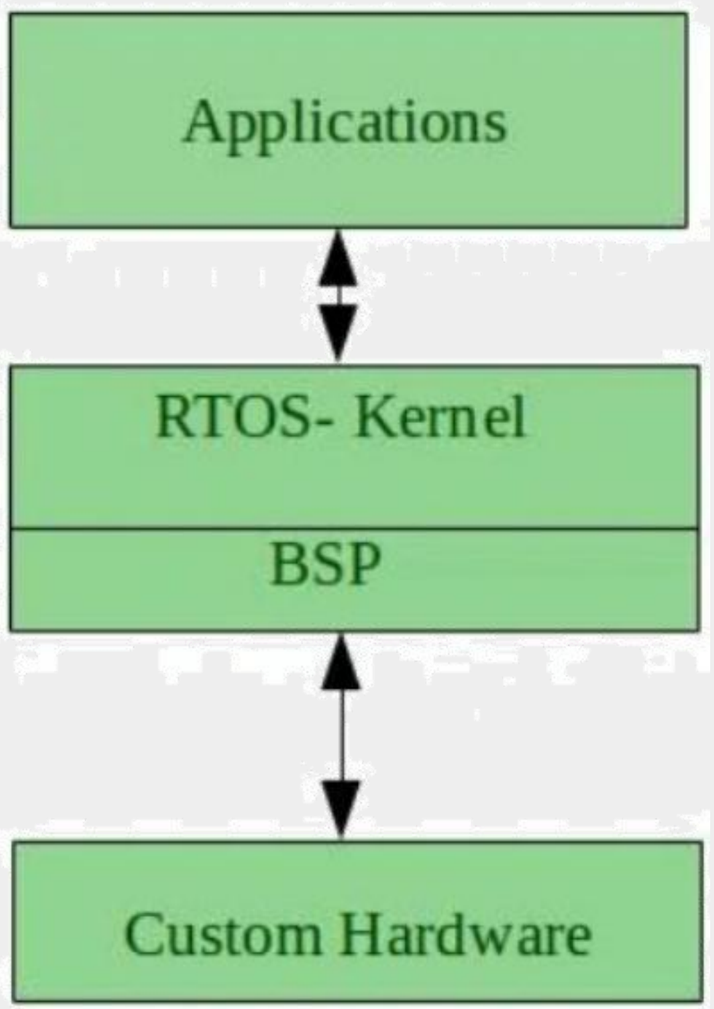
*Client-Server Communication client to request, server will return the response



Advantages	Disadvantages
Centralized servers provide high stability.	They require regular updates and maintenance.
Security issues are easier to handle through the servers.	Servers are expensive to buy and maintain.
It's easy to upgrade and integrate new technologies.	Users' reliance on a central server might be detrimental to workflows.

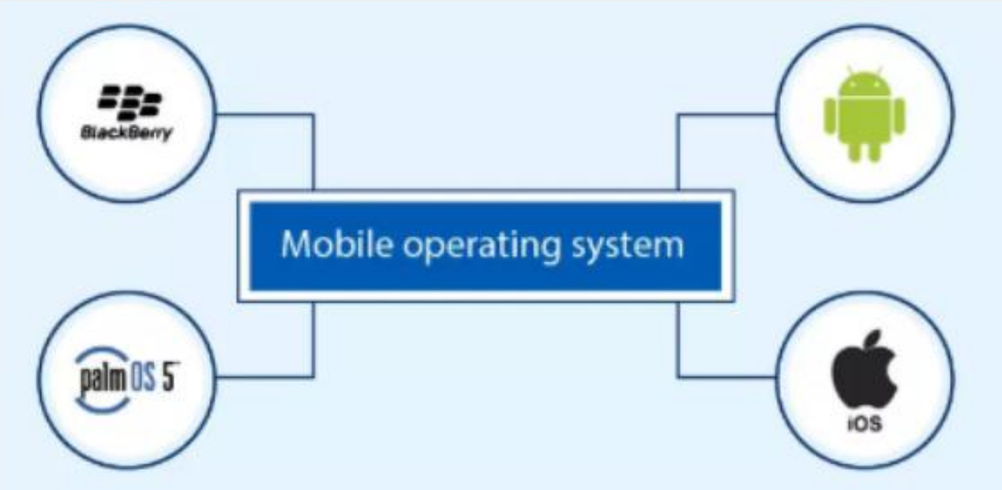
e. Real-time OS

- These types of OSs serve real-time systems.
- The time interval required to process and respond to inputs is very small.
- This time interval is called response time.
- Example with time requirements that are very strict like missile systems, air traffic control systems, robotics, computer vision, etc.
- Types of RTOS : Hard RTOS (strict), Soft RTOS(less strict)


Applications
RTOS- Kernel
BSP
Custom Hardware

Advantages	Disadvantages
They use devices and systems maximally, hence more output.	They have a low capacity to run tasks simultaneously.
They allow fast shifting from one task to another.	They use heavy system resources.
The focus is on current tasks, and less focus is put on the queue.	They run on complex algorithms that take work to understand.

e. Mobile OS



- Mobile os such as Android OS, Apple and Windows mobile OS run exclusively on small devices.
- Also manage wireless network connectivity.
- This combines the features of a personal computer with additional features useful for a handheld device.

Advantages	Disadvantages
Most systems are easy for users to learn and operate.	Some mobile OS put a heavy drain on a device's battery, requiring frequent recharging.
	Some systems are not user-friendly.

- OS come in various types, each used for specific needs. Whether it’s managing large batches of jobs, enabling multiple users to work simultaneously, coordinating networked computers, or ensuring timely execution in critical systems.
- Understanding these types helps in choosing the right operating system for the right job, ensuring efficiency and effectiveness.

Summary

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THANK YOU



